PROJECT MANUAL · VIVA & RESEARCH REFERENCE

Parkinson's Assessment & Monitoring

Voice-based Parkinson's screening research portal: setup, architecture, features, models, datasets, results, limitations and viva preparation.

SAKEC, Mumbai, India

Generated 15 October 2026

Research and screening support only — not a medical device or diagnosis.

This manual covers how to run the project, how it works, every model and dataset, all measured results, the limitations, and viva and research-paper preparation. Every number below comes from a metrics file in `models/` or from an experiment run in this project. The source is given next to each figure.

This is the single complete reference for the project.

How to read it

- **New to the project?** Read §1 (quick facts), §2 (problem), §5 (features) and §3 (how to run), then try the app.
- **Project member preparing for the viva?** Read everything; focus on §9 (models), §11 (all numbers), §17 (limitations) and §18 (questions and answers).
- **Writing the research paper?** Use §10 (datasets), §11 (result tables) and §19 (structure and safe claims).

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1. Quick facts (one-page revision sheet)

Item	Value
What it is	Web portal: record a sustained "aaah" → automatic voice measurements → Parkinson's screening result + UPDRS severity estimate → formal report/PDF → progress tracking → find care in India
Not	A medical device or diagnosis
Frontend	Next.js 13.5.6 (React 18.2, TypeScript), Recharts, Leaflet, @react-pdf/renderer
Backend	FastAPI (Python), SQLite, Praat via `praat-parselmouth` 0.4.7, scikit-learn 1.6.1, PyTorch 2.8
Ports	Frontend `http://localhost:3001`, API `http://127.0.0.1:8001` (`/docs` for OpenAPI)
Start	`./start.command` (macOS) or `start.bat` (Windows)
Screening for **recorded audio**	`rf-spectral-pd-screen-v4`: Random forest on 28 spectral, cepstral and pitch measures, trained on 78 phone recordings with age balancing. **ROC AUC 0.782** (0.765 on the age-matched ≥ 55 subgroup)
Screening for **typed/dataset values**	`logistic-pd-screen-v3`: Logistic regression on 13 log-scaled Praat measures, trained on 951 recordings from 284 speakers. **ROC AUC 0.758**; cross-dataset 0.831 / 0.732
Severity	`ridge-log-updrs-v2`: Ridge regression on 13 log-scaled Praat measures, trained on 5,875 recordings from 42 patients. **MAE 6.95** (cohort-mean MAE 7.13, R^2 −0.005)
Voice + gait	`dual-branch-fusion-updrs-v1`: CNN + MLP. **R^2 0.914 is leakage**; held-out-subject R^2 is **0.25 on average** (−0.12 to 0.72)
Result tiers (audio)	Not detected < 44% ≤ Inconclusive < 61% ≤ Detected
Audio tier error rates	Detected: 55% of PD, 13% of healthy · Not detected: 53% of healthy, 15% of PD
Key discovery	On phone audio, the original lab-style model scored **AUC 0.475** (chance) and flagged 72% of healthy people. Jitter and shimmer carry no signal on phone audio, while spectral and cepstral features do.
Datasets	Little et al. (195 rec, 32 spk) · Tsanas et al. (5,875 rec, 42 pts) · Sakar et al. (756 rec, 252 spk) · Iyer et al. (81 phone WAVs) · PhysioNet Gait in PD (gait signals)
Care finder	OpenStreetMap Nominatim (primary, India-only, healthcare types filtered) + Overpass (fallback); ~4–7 s per search
Tests	31 backend (pytest) + 21 frontend (Vitest), all passing

2. Problem statement and objectives

Problem. Parkinson's disease (PD) affects voice early. Typical changes are reduced loudness, breathiness, monotone pitch and irregular vocal-fold vibration (dysphonia/hypokinetic dysarthria). Neurologist access in India is uneven. A low-cost tool that a person can run on a phone or laptop could prompt earlier specialist consultation.

Objectives

1. Capture a short voice sample in the browser and measure clinically studied voice features automatically.

2. Screen for voice patterns associated with PD, and estimate motor severity (UPDRS Part III), with honest uncertainty.
3. Optionally fuse voice with gait (foot-pressure sensor) data.
4. Produce a formal, shareable report, track change over time, and connect users to care in India.
5. Be transparent: never show a score that was not validly computed, and state model accuracy openly.

Scope boundary. Research and screening support only. It is not diagnostic and has no clinical validation.

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3. How to run the project

3.1 Prerequisites

Tool	Version used	Notes
Python	3.9.6 (3.11+ recommended)	Virtual environment at `.venv/`
Node.js / npm	Node 24 (≥ 20.9 required)	
ffmpeg	**Not required**	The browser converts audio to WAV before upload
Internet	Required for Google sign-in, map tiles, care search, Google Fonts	

3.2 One-command start (macOS)

```
cd Capstone_Project
./start.command
```

What it does:

1. Selects `.venv/bin/python`.
2. Installs backend requirements.
3. Reads `NEXT_PUBLIC_GOOGLE_CLIENT_ID` from `frontend/.env.local`.
4. Starts the API on port 8001 with **auto-reload** (`--reload --reload-dir backend --reload-dir scripts`).
5. Starts Next.js on port 3001.
6. Opens **http://localhost:3001**.

Always use `localhost`, not `127.0.0.1`. Google sign-in only works on registered origins.

If a port is busy, stop the old instance (Ctrl + C) first. The launcher refuses to start twice.

Windows: run `start.bat` (fixed to run `backend.app:app` from the repository root).

3.3 Manual start

```
# Terminal 1 — API
python3 -m venv .venv && source .venv/bin/activate
pip install -r backend/requirements.txt
export FRONTEND_ORIGINS="http://localhost:3001,http://127.0.0.1:3001"
export GOOGLE_CLIENT_ID="<your web client id>"      # optional
uvicorn backend.app:app --host 127.0.0.1 --port 8001 --reload --reload-dir backend --reload-dir scripts

# Terminal 2 — frontend
cd frontend
npm install
# frontend/.env.local:  NEXT_PUBLIC_API_URL=http://127.0.0.1:8001
#                       NEXT_PUBLIC_GOOGLE_CLIENT_ID=<same client id>
npm run dev        # serves on port 3001
```

3.4 Environment variables

Variable	Where	Purpose / default
`NEXT_PUBLIC_API_URL`	frontend	API base URL (default `http://127.0.0.1:8001`)
`NEXT_PUBLIC_GOOGLE_CLIENT_ID`	frontend	Google Identity Services client ID
`GOOGLE_CLIENT_ID`	API	Must match the frontend ID; used to verify Google tokens
`FRONTEND_ORIGINS`	API	CORS allow-list (comma-separated)
`APP_DATA_DIR` / `APP_UPLOAD_DIR`	API	SQLite DB (`backend/.data/portal.sqlite3`) and uploads (`backend/uploads/`)
`MODEL_ARTIFACTS_DIR`	API	Model files (default `models/`)
`DIRECTORY_COUNTRY_CODES`	API	Care-finder country filter (default `in`; empty = worldwide)
`OSM_USER_AGENT`, `OSM_CONTACT_EMAIL`	API	Identify the app to OpenStreetMap services
`NOMINATIM_URL`, `OVERPASS_URL`	API	Override OSM endpoints

3.5 Google sign-in setup

1. Google Cloud Console → APIs & Services → Credentials → OAuth client (Web application).
2. **Authorized JavaScript origins:** add `http://localhost:3001`. No redirect URI is needed (popup flow).
3. Put the client ID in `frontend/.env.local` (`NEXT_PUBLIC_GOOGLE_CLIENT_ID`). `start.command` passes it to the API.
4. If the site is opened at `127.0.0.1`, the sign-in page shows a notice with a link to the localhost address.

3.6 Re-training the models (reproducibility)

```
source .venv/bin/activate
python -m scripts.train_vocal              # UPDRS ridge regressor  ->
models/updrs_regressor.pkl (+ metrics json)
python -m scripts.train_classifier         # lab-feature screening  ->
models/pd_classifier.pkl
python -m scripts.extract_audio_dataset    # extract features from data/iyer2023 WAVs ->
data/iyer2023_features.csv
python -m scripts.train_audio_classifier   # audio screening        ->
models/pd_audio_classifier.pkl
python -m scripts.train_fusion             # voice+gait fusion (research; see leakage notes)
python -m scripts.evaluate_fusion_leakage  # random vs subject-held-out fusion evaluation
```

Each trainer writes a `*_metrics.json` file (except fusion) containing the protocol, metrics, thresholds, SHA-256 hash of the artifact and limitations.

3.7 Running tests

```
.venv/bin/python -m pytest backend/tests -q     # 31 tests
cd frontend && npx vitest run                    # 21 tests
cd frontend && npx tsc --noEmit                  # type check
```

3.8 Troubleshooting

Symptom	Cause / fix
"Screening classifier could not be loaded"	The API was started before the models were retrained. Restart it; with auto-reload on, this won't recur.
Google stuck on "Loading…" / 403 "origin not allowed"	Open `http://localhost:3001` ; add that origin in Google Cloud Console
Recording rejected ("only X s of steady voice" / "dB above background")	Quiet room, ~20 cm from mic, one steady "aaah" for 5–10 s; use the Recording guide's microphone check
Care search slow or empty	OSM coverage varies; try a nearby city, larger radius, or "Hospitals & clinics"

4. System architecture and technology stack

```
Browser (Next.js 13 / React 18)
├─ Microphone (MediaRecorder) or file upload
│    └─ Web Audio API: decode → mono → resample 16 kHz → 16-bit PCM WAV   (lib/audio.ts)
├─ Pages: Home, Dashboard, New assessment, My assessments, Reports, Progress,
│         Find care (Leaflet map), Recording guide (live mic check), How it works …
└─ PDF generation in-browser (@react-pdf/renderer)
           │  HTTPS JSON / multipart, Bearer token
           ▼
FastAPI backend (backend/app.py)
├─ Auth: email+password (PBKDF2-SHA256), Google ID-token verification, sessions
├─ Assessments: consent, patient details, uploads, quality gates, feature extraction
│    └─ backend/audio_features.py  (Praat/Parselmouth + SciPy)
├─ Inference (scripts/inference.py)
│    ├─ Audio screening  : rf-spectral-pd-screen-v4   (models/pd_audio_classifier.pkl)
│    ├─ Lab screening    : logistic-pd-screen-v3      (models/pd_classifier.pkl)
│    ├─ Severity (voice) : ridge-log-updrs-v2         (models/updrs_regressor.pkl)
│    └─ Severity (fusion): dual-branch-fusion-updrs-v1 (models/fusion_model.pt + scaler)
├─ SQLite (backend/.data/portal.sqlite3) + uploads (backend/uploads/<user>/)
└─ Care finder (backend/specialists.py) → OpenStreetMap Nominatim / Overpass
```

4.1 Technology stack

Layer	Technology (version)	Why
UI framework	Next.js 13.5.6 App Router, React 18.2, TypeScript 5	Routing, server rendering of public pages
Fonts	Plus Jakarta Sans (headings), Inter (body) via `next/font/google`	Professional, readable typography
Icons	lucide-react 0.468	Consistent icon set
Charts	Recharts 3	Progress charts
Maps	Leaflet 1.9.4 + react-leaflet 4.2.1, OSM tiles	Interactive care map
PDF	@react-pdf/renderer 4.9	Client-side formal PDF
API	FastAPI 0.128 + Uvicorn	Typed REST API, OpenAPI docs
Voice analysis	praat-parselmouth 0.4.7 (Praat), SciPy 1.13, NumPy 2.0	Gold-standard phonetics measures
ML	scikit-learn 1.6.1, PyTorch 2.8, joblib	Classifiers/regressors, fusion CNN
Storage	SQLite with numbered migrations	Zero-config local database
Tests	pytest, Vitest + Testing Library, Playwright (manual E2E)	Automated verification

4.2 Repository layout

```
backend/            FastAPI app, audio_features.py, specialists.py, migrations/, tests/
scripts/            training, inference, feature contract, dataset tools
models/             trained artifacts + *_metrics.json
data/               datasets (see §10)
frontend/app/       pages (Next.js App Router)
frontend/components UI components (wizard, report, map, illustrations…)
frontend/lib/       api client, audio conversion, report helpers, site config
docs/               this manual
start.command / start.bat
```

5. Features, page by page

Page (route)	What it does
Home (`/`)	Landing page with animated voice-analysis card (live waveform, feature chips, self-drawing trend line), feature sections, scroll-reveal animations
Register / Sign in (`/register` , `/signin`)	Email + password accounts; Google sign-in (popup); session tokens valid 30 days
Dashboard / Overview (`/dashboard`)	Welcome hero with illustration; stats (assessments, reports, latest date, **latest screening verdict**); quick-action tiles; recent assessment cards
New assessment (`/assessment/new`)	4-step wizard: ① patient details (name required; age, sex, location) + consent; ② type: voice or voice + gait; ③ voice input via **Record** (mic button, live level meter, "too quiet" warning), **Upload audio** (any browser-decodable format), or **Feature values** (16 fields or load `parkinsons.data` rows); ④ gait file (19-column, ≥ 200 rows) for fusion
Report (`/reports/[id]`)	Opens automatically after a successful submission with a "report is ready" banner. Formal report: patient info, assessment details, screening verdict with probability bar and tier zones, UPDRS gauge with ± error, acoustic and voice-quality tables, limitations. PDF download; delete
Rejected recording	If quality gates fail: clear reason, recording tips, **Record again** (keeps patient details), link to guide
My assessments (`/assessments`)	Summary strip, filters (All / With report / Need attention), cards with colour-coded tier, probability, UPDRS, patient; **Delete with inline confirmation** (removes record + audio file)
Assessment record (`/assessments/[id]`)	Raw record view, status, files, record again, delete
Reports (`/reports`)	All completed reports as cards; illustrated empty state
Progress (`/progress`)	Screening probability over time on coloured tier bands; selectable voice-measure trend (CPPS, pitch SD, HNR, jitter, shimmer); summary cards. Replaces the old "Analytics" page (`/analytics` redirects)
Find care (`/specialists`)	India-only search by city/area/PIN, quick city chips, care-type and radius filters, **interactive Leaflet map** with colour-coded markers synced to result list, Directions (Google Maps), OSM links, national centres panel (PDMDS, NIMHANS, AIIMS, WHO)
Recording guide (`/guide`)	Animated 4-step tutorial carousel (auto-play with progress dots) + **live microphone check** (too quiet / good / too loud), do's and don'ts

Page (route)	What it does
How it works (`/how-it-works`)	Animated 6-step pipeline, meaning of each result tier, accuracy figures
About / Resources	Project approach and curated resources
Footer	Brand, "SAKEC, Mumbai, India", platform/learn links, India helplines (Emergency 112, Tele-MANAS 14416), PDMDS & NIMHANS links, disclaimer

6. What was added or changed in the latest development phase

6.1 Before → after summary

Area	Before	After
Browser recordings	Failed: WebM needed ffmpeg, which was not installed	Browser converts any audio to 16 kHz mono WAV; ffmpeg not needed
Feature extraction bugs	Jitter(%) ×100 too large; NHR computed as 1/HNR(dB)	Jitter stored as fraction (matches datasets); NHR = 10^(–HNR/10)
Recording quality	Whole clip analysed, including silence and noise → false "Detected"	Only the longest steady voiced segment is analysed; recordings rejected if < 1.5 s steady voice or SNR < 15 dB
Screening	None (UPDRS estimate only)	Two screening models (audio-specific and lab-feature) with **three tiers** and published error rates
Screening accuracy on phone audio	AUC 0.475; 72% of healthy flagged	AUC 0.782; 13% of healthy flagged
Severity model	Random forest, 16 features, R^2 –0.105	Ridge on 13 log features, MAE 6.95; shown only when screening = Detected, with ± MAE
Report	Key-value dump	Formal report: patient details, verdict, gauge, tables, limitations; formal PDF
Submission journey	Inline result	Redirect to report page with success banner; rejected recordings get tips + Record again
My assessments	Plain list, no delete	Cards, stats, filters, delete with confirmation (DB + files)
Analytics	Always empty (depended on non-existent fields)	Progress page with real trends
Care finder	Overpass mostly down → junk name matches ("Doctor … Chowk" road), US placeholder, 250 km default	Nominatim POI search filtered to real healthcare types, India-only, relevance ranking, map, 25 km default, Indian examples
Google sign-in	Stuck on "Loading"; origin 403	Script readiness fixed; launcher opens localhost; origin notice
Branding/UI	Letter "P" logo, unloaded fonts	Waveform-shield logo + favicon, Plus Jakarta Sans/Inter, professional dark footer, animations, illustrations, guide
Ops	Stale servers after code changes	API auto-reload; Windows launcher fixed

6.2 New files (selection)

- **Backend/scripts:** `scripts/train_classifier.py`, `scripts/train_audio_classifier.py`, `scripts/extract_audio_dataset.py`, `scripts/evaluate_fusion_leakage.py`, `backend/migrations/005_assessment_patient_details.sql`
- **Data:** `data/pd_speech_features.csv`, `data/iyer2023/` (WAVs + demographics), `data/iyer2023_features.csv`
- **Models:** `pd_classifier.pkl`, `pd_audio_classifier.pkl`, `updrs_regressor.pkl` + metrics JSON
- **Frontend:** `lib/audio.ts`, `lib/report.ts`, `lib/site.ts`, `components/severity-gauge.tsx`, `brand-logo.tsx`, `site-footer.tsx`, `illustrations.tsx`, `care-map.tsx`, `mic-check.tsx`, `progress-view.tsx`; pages `/guide`, `/progress`; `app/icon.svg`, `app/template.tsx`

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7. Audio processing pipeline

7.1 In the browser (`frontend/lib/audio.ts`)

1. **Capture.** `MediaRecorder` records with echo cancellation, noise suppression and auto-gain **off**, so the voice signal is not altered. Max 60 s. A live RMS meter warns if the level stays below 0.02 (about –34 dBFS).
2. **Normalise.** `AudioContext.decodeAudioData` decodes the audio, then `OfflineAudioContext(1 channel, 16 kHz)` down-mixes and resamples it. It is encoded as 16-bit PCM WAV and the duration is read from the decoded buffer.
3. **Upload** with consent, patient details and modality.

7.2 On the server (`backend/audio_features.py` , version `audio-extractor-v4`)

Step	Rule
Decode	WAV in-process (ffmpeg only needed for non-WAV uploads that bypass the browser converter)
Basic quality	≥ 3 s, finite, not silent (RMS > 1e-5), clipped fraction ≤ 1%, duration matches the browser report (± 2 s)
Steady segment	Praat pitch track (10 ms, 75–500 Hz); longest run of voiced frames whose f0 changes ≤ 15% frame-to-frame; trim 0.15 s at each end; needs ≥ **1.5 s** steady voice after trimming
Signal-to-noise	Voice RMS vs median RMS of unvoiced frames; reject if **< 15 dB** (skipped if fewer than 30 unvoiced frames exist)
16 contract features	Praat jitter/shimmer/HNR/NHR + in-repo RPDE/DFA/PPE (best-effort; null if unidentifiable)
28 spectral features	Segment band-limited to 0–4 kHz (8 kHz sampling, telephone band) then MFCC, CPPS, pitch variability, spectral tilt
Store	Features, quality info (voiced fraction, steady seconds, SNR, median f0, analysis window)

Why band-limit to 4 kHz? The audio model was trained on telephone recordings (8 kHz). Measuring every input in the same band keeps live recordings comparable to the training data, whether they come from a laptop, phone or studio microphone.

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8. Voice features explained

8.1 The 16-feature "contract" (`scripts/feature_contract.py`)

Feature	Meaning	Praat call
Jitter(%)	Cycle-to-cycle variation of pitch period, relative (stored as a fraction)	Get jitter (local)
Jitter(Abs)	Absolute period variation (seconds)	jitter (local, absolute)
Jitter:RAP	Relative average perturbation (3-period)	jitter (rap)
Jitter:PPQ5	5-point period perturbation quotient	jitter (ppq5)
Jitter:DDP	Average absolute difference of differences (= 3 × RAP)	jitter (ddp)
Shimmer	Cycle-to-cycle amplitude variation	shimmer (local)
Shimmer(dB)	Same in decibels	shimmer (local_dB)
Shimmer:APQ3/APQ5/APQ11	3/5/11-point amplitude perturbation quotients	shimmer (apq3/5/11)
Shimmer:DDA	= 3 × APQ3	shimmer (dda)
NHR	Noise-to-harmonics ratio = 10^(–HNR/10)	derived
HNR	Harmonics-to-noise ratio (dB)	Harmonicity (cc) mean
RPDE	Recurrence period density entropy (in-repo implementation)	SciPy
DFA	Detrended fluctuation analysis exponent (in-repo)	SciPy
PPE	Pitch period entropy (in-repo)	SciPy

Important: the in-repo RPDE/DFA/PPE are **not calibrated** to the original datasets. Extracted PPE (~0.61–0.67) sits above the training maximum (0.527). That alone made every recording screen positive, so **no deployed model uses RPDE, DFA or PPE**. They are stored for transparency only.

8.2 The 28 spectral features (`SPECTRAL_FEATURE_COLUMNS`), used by the audio screening model

Group	Count	Meaning
MFCC1–12 SD	12	Variability of each mel-frequency cepstral coefficient across the vowel (timbre stability)
MFCC1–12 delta SD	12	Variability of frame-to-frame change (articulatory steadiness)
CPPS	1	Smoothed cepstral peak prominence (dB): a robust measure of dysphonia/breathiness; lower values mean less periodic voice
F0 SD (semitones)	1	Pitch steadiness while sustaining
F0 range (semitones)	1	5th–95th percentile pitch spread
Spectral tilt (dB)	1	Mean LTAS energy 0–1 kHz minus 1–4 kHz

Observed medians in phone data (Iyer): CPPS **12.57 dB healthy vs 10.27 dB PD**; F0 SD **0.160 vs 0.241 semitones**.

CPPS uses Praat's "Robust" trend fit. It is about 50× faster than "Robust slow" (0.07 s vs 3.99 s per clip) with near-identical values.

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9. Models (model cards)

9.1 Audio screening — `rf-spectral-pd-screen-v4` (used for recorded/uploaded audio)

Field	Value
File	`models/pd_audio_classifier.pkl` ; metrics `pd_audio_classifier_metrics.json`
Algorithm	RandomForestClassifier (500 trees, min_samples_leaf 4)
Inputs	28 spectral features (§8.2), clipped to the training range
Training data	Iyer et al. 2023 phone recordings: 78 usable (39 PD, 39 healthy) of 81
Confound handling	**Age-balancing inverse-probability weights** per decade bin (healthy median age 44 vs PD 67.9). Bins containing only one class are down-weighted to 0.05
Validation	StratifiedKFold 5 folds × 20 repeats (one recording per person, so speaker-independent)
ROC AUC	**0.782**
ROC AUC, age ≥ 55 subgroup	**0.765**
Corr(score, age) in healthy	**−0.029** (score does not track age)
Thresholds	Not detected < **0.44** ≤ Inconclusive < **0.61** ≤ Detected (chosen so ≤ 15% healthy are "Detected" and ≤ 15% PD are "Not detected")
Tier rates (people with PD)	Detected **55.1%**, Inconclusive 29.9%, Not detected **15.0%**
Tier rates (healthy)	Detected **13.1%**, Inconclusive 33.5%, Not detected **53.5%**
SHA-256	`ee60780a…952c`

9.2 Lab-feature screening — `logistic-pd-screen-v3` (used for typed/dataset feature values)

Field	Value
File	`models/pd_classifier.pkl`
Algorithm	Pipeline: log(12 perturbation features) → StandardScaler → LogisticRegression (L2, C = 0.3, class_weight balanced)
Inputs	13 Praat measures (Jitter ×5, Shimmer ×6, NHR, HNR); clipped to training range
Training data	Sakar 2019 (756 rec, 252 speakers) + Little (195 rec, 32 speakers) = **951 recordings, 284 speakers (212 PD, 72 healthy)**
Validation	StratifiedGroupKFold by speaker, 5 folds × 10 repeats; cross-dataset transfer

Field	Value
ROC AUC (pooled CV)	**0.758**
Train Sakar → test Little	**0.831**
Train Little → test Sakar	**0.732**
Thresholds	Not detected < **0.35** ≤ Inconclusive < **0.70** ≤ Detected
Tier rates (PD)	Detected 38.3%, Inconclusive 49.3%, Not detected 12.4%
Tier rates (healthy)	Detected 11.5%, Inconclusive 41.8%, Not detected 46.8%
Caveat	On phone audio processed by this app it scores **AUC 0.475**, which is why recorded audio uses model 9.1

9.3 Severity — `ridge-log-updrs-v2`

Field	Value
File	`models/updrs_regressor.pkl`
Algorithm	log features → StandardScaler → Ridge (α = 10)
Target	motor UPDRS (original UPDRS Part III; dataset range 5.0–39.5, mean 21.3)
Data	Tsanas telemonitoring: 5,875 recordings, 42 people with early PD
Validation	GroupKFold by subject (6 folds)
Metrics	**MAE 6.95**, RMSE 8.15, R^2 –0.005; predicting the cohort mean gives MAE 7.13
Interpretation	Not better than the cohort average for new people, so shown as "indicative ± 7" and **only when screening = Detected**

9.4 Voice + gait fusion — `dual-branch-fusion-updrs-v1` (research)

Field	Value
Files	`models/fusion_model.pt` , `models/fusion_vocal_scaler.joblib`
Architecture	Vocal MLP (16→32→16) + dual-branch 1-D CNN, one branch per foot (8 sensors each; Conv1d 8→16→32, max-pool, global avg-pool, FC 64→32) → fusion head (48→24→1, dropout 0.2)
Training	Adam lr 1e-3, StepLR (20, 0.5), 60 epochs, MSE, batch 64
Gait input	First 200 rows of the 19-column PhysioNet format (time, 8 left, 8 right, 2 totals)
Pairing	`dataset_alignment.py` : nearest neighbour on [age, motor UPDRS]; **42 voice subjects → 27 gait patients** (mean match distance 2.04)
Claimed result	R^2 0.90 / RMSE 2.52 (hard-coded in `plot_comparison.py`)
Reproduced, random split	**R^2 0.914, RMSE 2.40, MAE 1.82** (mean-guess MAE 7.05)
Subject-held-out (3 seeds)	**R^2 0.248 on average** (–0.119, 0.143, 0.719), MAE **4.45**, RMSE 5.78; mean-guess MAE 7.23

Field	Value
Verdict	The 0.90 figure is **data leakage** from the same subjects in train and test. Even the held-out result is inflated, because gait files were chosen using the target (UPDRS), and voice and gait come from **different people**. Keep it as a research demonstration only.

9.5 Legacy baseline — `random-forest-updrs-v1`

Random forest on 16 features, subject-held-out 80/20 (33 train / 9 test subjects): **RMSE 8.03, MAE 6.77, R^2 –0.105**. It was replaced by 9.3. The earlier hard-coded "baseline R^2 0.33" was also from a non-subject-independent protocol.

9.6 Other research scripts in the repo (not used by the portal)

- `scripts/train_cnn.py`: drawing (spiral/wave) image classifiers (VGG16/ResNet50/DenseNet121/EfficientNetV2S); data in `data/drawings/` (102 spiral, 102 wave images).
- `scripts/fusion_engine.py`, `scripts/explain_model.py`: class-level meta-learner and SHAP explanation. They align datasets by class, not by person, so they are not subject-level multimodal evidence.
- `main.py`: terminal menu that runs alignment, baseline, fusion and the comparison chart.

10. Datasets

Dataset	Source / citation	Size	Labels	Used for
Parkinsons (`data/parkinsons.data`)	Little et al. 2007/2009; UCI ML Repository	195 sustained /a/ recordings, **32 speakers** (24 PD, 8 healthy; 147/48 recordings)	status (1 = PD)	Lab screening model; manual "load dataset" feature
Parkinsons Telemonitoring (`data/parkinsons_updrs.data`)	Tsanas et al. 2010; UCI	**5,875 recordings, 42 people** with early PD (28 male / 14 female), age 36–85	motor_UPDRS (5.0–39.5), total_UPDRS (7–55)	Severity model; fusion model

Dataset	Source / citation	Size	Labels	Used for
PD Speech Features (`data/pd_speech_features.csv`)	Sakar et al. 2019; UCI #470, **CC BY 4.0**	756 recordings, **252 speakers** (188 PD, 64 healthy), 3 per speaker; 754 features	class	Lab screening model (13 Praat columns)
Voice samples PD/HC (`data/iyer2023/`)	Iyer et al. 2023; figshare doi 10.6084/m9.figshare.23849127, **CC BY 4.0**	81 phone WAVs, 8 kHz, ~1.5–7.2 s (40 PD, 41 HC); 78 pass quality gates	PD/HC, age, sex	**Audio screening model**; external validation
Gait in Parkinson's Disease (`data/gait-in-parkinsons-disease-1.0.0/`)	PhysioNet (Hausdorff et al.; sub-studies Ga/Ju/Si)	306 gait recordings: GaPt 75, GaCo 38, JuPt 104, JuCo 25, SiPt 35, SiCo 29	Pt = PD, Co = control	Fusion model; gait upload validation
Drawings (`data/drawings/`)	Spiral/wave drawing dataset	102 spiral + 102 wave images	healthy / parkinson	CNN research script only
Alignment map (`data/aligned_patients_mapping.csv`)	Generated	42 rows	–	Synthetic voice↔gait pairing

Gait file naming: `GaPt03_01.txt` = Ga study, **Pt**(PD) subject 03, walk 01; `Co` = control. 19 columns at 100 Hz: time, 8 left-foot sensors, 8 right-foot sensors, total left, total right (vertical ground-reaction force, N).

Iyer dataset demographics (78 usable): healthy median age **44**, PD median **67.9**. Sex: healthy 25 F / 14 M, PD 18 F / 21 M. Of the 3 excluded clips, 2 had low SNR (12 dB, 6 dB) and 1 had only 0.7 s of steady voice.

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11. Experiments and findings log (all numbers)

11.1 Bugs found in the original audio pipeline

Finding	Evidence	Fix
Browser recordings never scored	"WEBM decoding is unavailable because ffmpeg is not installed"	Browser-side WAV conversion

Finding	Evidence	Fix
Jitter(%) 100× too large	Datasets store a fraction (median 0.0049); extractor multiplied by 100	Removed ×100
NHR wrong formula	1/HNR(dB); correct linear relation 10^(–HNR/10) correlates **0.93** with dataset NHR	Formula fixed
False "Detected" on a healthy user	Recording only **22% voiced**, longest steady run **0.3 s**; voice RMS ~0.008 vs noise ~0.0035; jitter 3.0%, shimmer 0.21, HNR 5.7 dB; model said 99.99%	Steady-segment selection + SNR/duration rejection
Valid recordings rejected	7 of 81 Iyer clips failed only on RPDE/DFA/PPE range checks	Nonlinear features made best-effort

11.2 Screening on Little data only (32 speakers, speaker-grouped CV, 5 repeats)

Columns: AUC / balanced accuracy / sensitivity / specificity

Model	16 features	13 Praat features
Logistic regression (balanced)	0.757 / 0.636 / 0.731 / 0.542	**0.743 / 0.682 / 0.668 / 0.696**
Logistic, log features	0.742 / 0.667 / 0.796 / 0.538	–
SVM (RBF)	0.653 / 0.570 / 0.852 / 0.288	–
Random forest	0.689 / 0.586 / 0.850 / 0.321	0.624 / 0.526 / 0.810 / 0.242
Extra trees	0.702 / 0.603 / 0.819 / 0.388	–

The first prototype (RF, single CV) scored accuracy 0.713, AUC 0.703, **sensitivity 0.864 but specificity 0.25** (it flagged 3 of 4 healthy speakers). That is why it was replaced.

11.3 Adding Sakar et al. (AUC; speaker-grouped CV)

Model	CV Sakar	CV Little	CV combined	Sakar→Little	Little→Sakar
Logistic	0.736	0.743	0.736	0.807	0.680
Logistic, log features	**0.754**	0.697	**0.757**	**0.831**	**0.732**
SVM, log	0.718	0.569	0.724	0.776	0.700
Random forest	0.697	0.627	0.689	0.666	0.697
Extra trees	0.706	0.671	0.707	0.776	0.684
Gradient boosting	0.711	0.566	0.702	0.738	0.694

Regularisation sweep (log-LR, combined CV / S→L / L→S): C 0.03 → 0.738/0.843/0.711; C 0.1 → 0.750/0.843/0.721; **C 0.3 → 0.756/0.831/0.732 (chosen)**; C 1 → 0.759/0.804/0.741; C 3 → 0.759/0.778/0.745. L1 and dropping the redundant DDP/DDA columns made no difference.

Threshold study (combined, 20 repeats):

Single cut-off	Sensitivity	Specificity
0.50	0.69	0.69
0.60	0.54	0.80

Single cut-off	Sensitivity	Specificity
0.70	0.39	0.88
0.75	0.30	0.92

Three-tier option chosen: **0.35 / 0.70**. Healthy flagged 12%, PD cleared 12%, about 47% inconclusive.

11.4 External validation on phone audio (Iyer, our extractor) — the key finding

Test	Result
Lab model v3 applied to phone recordings	**AUC 0.475** (chance); PD and healthy both 72% "Detected", median p = 0.81
Lab model retrained including phone data	AUC 0.496
Phone-only model, 13 Praat features	AUC 0.514
Medians healthy vs PD (phone)	Jitter 0.0044 vs 0.0041; Shimmer 0.054 vs 0.055; HNR 19.4 vs 19.1 dB, i.e. **no separation**

Feature sets on phone audio (AUC; LR / RF / SVM; 20 × 5-fold):

Feature set	LR	RF	SVM
Age only (confound check)	**0.873**	0.834	0.834
Age + sex	0.865	0.830	0.831
13 Praat	0.443	0.452	0.525
Spectral (all)	0.766	0.817	0.782
Spectral + Praat	0.770	0.824	0.779
CPPS + F0 + tilt	0.681	0.731	0.659
MFCC means only	0.483	0.600	0.402

Age-matched check (age ≥ 55: n = 45; 36 PD, 9 healthy; median ages 68.4 vs 68.0): age only **0.473**; spectral LR **0.662**, RF 0.656; 13 Praat 0.495 / 0.400. **Spectral features carry signal beyond age.**

Age-balanced training (AUC full / age ≥ 55 / corr(score, age) in healthy):

Feature set & model	Plain	Age-weighted
Spectral (all), RF	0.813 / 0.741 / +0.25	0.780 / 0.761 / +0.04
MFCC SD + delta SD + CPPS + F0 + tilt, RF	0.812 / 0.742 / +0.17	**0.790 / 0.772 / –0.02**
Same, LR C 0.03	0.777 / 0.824 / +0.06	0.729 / 0.770 / +0.01

Chosen: the bold row (age-weighted RF). Final trained version: AUC 0.782, age ≥ 55 0.765, corr –0.029.

11.5 Severity regressors (subject-grouped 6-fold, motor UPDRS)

Features	Model	MAE	R^2
—	Predict cohort mean	7.13	–0.047

Features	Model	MAE	R^2
16	Random forest	7.32	−0.149
16	Ridge (log)	7.03	−0.037
16	Gradient boosting	7.27	−0.113
13	**Ridge (log) — deployed**	**6.95**	**−0.005**
13	Random forest	7.23	−0.092
13 + age + sex	Ridge (log)	6.99	−0.019
13 + age + sex	Random forest	8.03	−0.481
Age + sex only	Gradient boosting	7.48	−0.281

Conclusion: with 42 patients, voice cannot predict UPDRS across people better than the mean.

11.6 Fusion leakage audit (`scripts/evaluate_fusion_leakage.py`)

Protocol	RMSE	MAE	R^2	Mean-guess MAE
Random 80/20 split (original)	2.40	1.82	**0.914**	7.05
Subject held-out, seed 0	5.99	4.18	−0.119	6.03
Subject held-out, seed 1	6.69	5.23	0.143	7.88
Subject held-out, seed 2	4.64	3.93	0.719	7.78
Subject held-out, mean	**5.78**	**4.45**	**0.248**	7.23

11.7 Sanity checks with synthetic voices (final models)

Synthetic voice	Audio-model probability	Tier
Clean (HNR 28 dB)	0.40–0.51	Not detected → Inconclusive
Mild noise (HNR 21 dB)	0.69	Inconclusive (lab model)
Rough (HNR 9.7 dB, shimmer 0.11)	0.87	Detected
Very rough (HNR 4.6 dB)	0.96	Detected

End-to-end check with repo files (lab model + fusion): PD row + `GaPt03_01` → **Detected (77%)**, UPDRS 18.9; healthy row + `GaCo03_01` → **Not detected (21%)** (fusion UPDRS 36.9 is hidden as not applicable).

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12. Severity scale (UPDRS) and result tiers

- **UPDRS Part III (motor examination)** scores 0–108. Higher means worse motor signs.
- Severity bands follow Martínez-Martín et al. (2015) MDS-UPDRS Part III cut-offs: **Mild ≤ 32, Moderate 33–58, Severe ≥ 59**. Caveat: the training data uses the *original* UPDRS, not the MDS-UPDRS, so the bands are approximate. They are **not Hoehn & Yahr stages**.
- The report gauge shows score ± MAE (± 7), labelled "indicative". It is greyed out unless screening = Detected, because the severity model was trained only on people with PD.

- **Tiers:** *Detected* = see a neurologist; *Inconclusive* = re-record or consult; *Not detected* = no voice markers found, which cannot rule PD out. Each report states the error rate for its tier, e.g. "53% of healthy and 15% of PD speakers received this result".

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13. Care finder (OpenStreetMap)

Aspect	Detail
Geocoding	Nominatim `/search` (jsonv2), `countrycodes=in`, accepts city, area or 6-digit PIN
Primary provider search	Nominatim POI search inside a bounding box of the chosen radius; up to 6 search terms per care type (e.g. neurology: "neurology", "neuro", "neurologist", "hospital")
Type filter	Keep only `amenity` = hospital/clinic/doctors/nursing_home/social_facility, any `healthcare=*`, `office` = ngo/charity/association/foundation/physician/therapist, or Parkinson-named non-road features. **Drops roads/buildings named after doctors**
Ranking	Names matching the specialty (neuro/brain/Parkinson/movement; physio/rehab) first, then distance
Fallback	Overpass API (`overpass-api.de`, then `overpass.kumi.systems`), 15 s timeout. `overpass.osm.ch` removed (Swiss data only)
Etiquette	User-Agent identifies the app; 1 request/s per service; results cached 15 min (128 entries); geocodes cached (LRU 256)
Speed (live test)	Mumbai neurology 32 results in 5.0 s; PIN 400001 hospitals 97 in 4.5 s; Pune physio 5 in 4.2 s; Bengaluru 63 in 6.7 s
UI	Leaflet map with OSM tiles, colour by type (hospital red, clinic blue, doctor purple, physio green, NGO amber), select-to-fly, Directions link to Google Maps, OSM link
Licensing	Data © OpenStreetMap contributors, ODbL; attribution shown on page and map
Privacy	Browser location is requested only on "Use my location" and sent only on Search

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14. Backend API, database and security

14.1 Endpoints

Method & path	Purpose
GET `/api/health`	Liveness
GET `/api/models/status`	Availability of voice, screening (lab + audio) and fusion artifacts
POST `/api/auth/register` · `/login` · `/google` · `/logout` ; GET `/api/auth/me`	Accounts and sessions
POST `/api/assessments`	Audio (+ optional gait) assessment: multipart with consent, modality, duration, voice, gait, patient_*
POST `/api/assessments/manual`	16 feature values (+ optional gait)
GET `/api/assessments` , `/api/assessments/{id}`	History (owner-scoped)

Method & path	Purpose
DELETE `/api/assessments/{id}`	Delete record + uploaded files (owner only; 404 otherwise)
GET `/api/reports`, `/api/reports/{id}`	Completed assessments
GET `/api/dashboard`	Counts + recent
POST `/api/research/predict-voice-features`	Raw UPDRS prediction for 16 values
GET `/api/specialists`	Care search (location, category, radius, optional lat/lon)

14.2 Database (SQLite, migrations 001–005)

- `users` (id, email unique, name, password_hash, password_salt, google_subject unique, created_at)
- `sessions` (token_hash PK, user_id → users, expires_at)
- `assessments` (id, user_id, modality, status ∈ {completed, validation_required, unavailable, …}, unavailable_reason, created_at, consent_version, consented_at, duration_seconds, files_json, **result_json**, **extracted_features_json**, **extraction_version**, **quality_json**, **patient_json**). Indexed by (user_id, created_at) and (user_id, status, created_at)

14.3 Security and privacy

- Passwords: **PBKDF2-HMAC-SHA256, 310,000 iterations**, random salt.
- Sessions: random bearer token; only its **SHA-256 hash** is stored; expires after **30 days**.
- Google: ID token verified server-side against `GOOGLE_CLIENT_ID`; accounts cannot be taken over via a different Google subject.
- Every assessment query is scoped by user_id; uploads are stored per user; deletion removes files.
- Uploads ≤ 25 MB; audio magic-byte checks (WAV/WebM/OGG/MP3/M4A); gait strict 19-column numeric validation.
- Consent is required and versioned (`consent_version` v1, timestamp stored).
- CORS restricted to `FRONTEND_ORIGINS`.

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15. Reports and PDF

Report number `PD-XXXXXXXX` (first 8 hex characters of the record ID). Sections:

1. Patient information
2. Assessment details: date, pathway, voice source, severity model, screening model
3. Findings: verdict card with probability bar over tier zones and tier error rates; UPDRS gauge with ± MAE and band legend; interpretation paragraph
4. Acoustic measurements (16 features + CPPS, F0 SD, F0 range, spectral tilt)
5. Notes and limitations

The PDF is generated in the browser (A4, Helvetica) with a vector gauge and page numbers.

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16. Testing and verification

Suite	Count	Covers
Backend pytest	**31**	Auth/sessions, Google sign-in, migrations (v5), audio extraction, quality rejection, patient validation, three-tier screening, audio-model routing, delete (owner-only, files removed), gait validation, specialist search (filtering, fallback, rate limit, caching, India codes)
Frontend Vitest	**21**	Navigation, auth guard, wizard validation, dataset loading, consent-gated mic, submit → redirect to report, rejected flow, report rendering (detected/inconclusive), progress empty state, delete confirmation, care search
Type check	`tsc --noEmit` clean	
Browser E2E (Playwright, manual)	Full journey	Home, sign-in (Google button after client navigation), audio + gait submissions, rejection, history, delete, progress, live Mumbai search (37 listings), guide mic check, analytics redirect: **0 console errors**

`tests/test_research_foundation.py` (root) has a pre-existing import error (`generate_comparison` missing from `plot_comparison.py`). Its voice-model tests were updated and pass when run in isolation.

17. Limitations and ethics

1. **Small, non-representative data.** The audio model uses 78 people from one dataset (one country, phone channel). There are no Indian-language or Indian-population recordings.
2. **Modest accuracy.** AUC ~0.78. About 1 in 8 healthy people get "Detected" and 15% of PD get "Not detected". It is a screening aid only.
3. **Age confound** in the training data. Mitigated by weighting and checked in an age-matched subgroup (n = 45, only 9 healthy, so the estimate is noisy).
4. **No clinical or external prospective validation.** Microphones and rooms differ from the training conditions.
5. **Severity estimate** is not better than the average for new people.
6. **Fusion** pairs different people via the target variable. It is not multimodal evidence.
7. **RPDE/DFA/PPE implementations** are not calibrated.
8. **Medication state, language and disease stage** are not modelled.
9. **Ethics.** Explicit consent, no diagnosis claims, error rates shown, data private to the account, deletion supported, helplines shown, and results never fabricated (failed analyses are saved without scores).

18. Viva preparation: likely questions and answers

Q: Why voice for Parkinson's? Hypokinetic dysarthria appears in up to ~90% of people with PD and can be early. It shows up as reduced loudness, breathiness, monotone pitch and irregular phonation. A sustained vowel isolates phonation from language, so it suits a language-independent tool for India.

Q: What is jitter / shimmer / HNR? Jitter is cycle-to-cycle variation in pitch period, shimmer in amplitude, and HNR is the harmonic energy relative to noise. They come from Praat on a steady vowel.

Q: What is CPPS and why did it matter? Cepstral peak prominence (smoothed) measures how clearly periodic the voice is. It is robust to recording conditions and a standard dysphonia marker. In the phone data it was 12.6 dB (healthy) vs 10.3 dB (PD).

Q: Why two screening models? Features from lab microphones (Little/Sakar) and from phone audio behave differently. The lab model scored AUC 0.475 on phone audio processed by the app. Typed lab values use the lab model; recordings use the audio model trained on audio processed by the same extractor.

Q: How did you avoid data leakage? All splits are by speaker/subject (StratifiedGroupKFold / GroupKFold). The Iyer data has one recording per person. Scalers sit inside pipelines fitted on training folds. I also showed that the inherited fusion result (R^2 0.91) was leakage: it falls to 0.25 when subjects are held out.

Q: Why three tiers instead of yes/no? A single threshold forces a trade-off: at 0.5 about 30% of healthy people are flagged. Tiers report only confident results and send borderline ones to re-recording or a clinician. Each tier's error rate is printed on the report.

Q: Why random forest for audio but logistic regression for lab features? For lab features, logistic regression on log features generalised best across datasets (0.831 / 0.732). For the 28 spectral features with age weighting, RF gave the best age-matched AUC with near-zero age correlation (0.772 / –0.02).

Q: How did you handle the age confound? Age alone gave AUC 0.873. I used inverse-probability age-balancing weights per decade, evaluated on the age ≥ 55 subgroup (0.765), and checked that scores do not correlate with age in healthy speakers (–0.029).

Q: Why is the UPDRS estimate shown only when Detected? The regressor was trained only on PD patients, so outputs for healthy voices are meaningless. In one test, a healthy profile produced 36.9 ("Moderate"). It is also no better than the mean (MAE 6.95 vs 7.13).

Q: What are AUC, sensitivity and specificity? AUC is the probability that a random PD case scores higher than a random healthy one (0.5 = chance, 1 = perfect). Sensitivity is the fraction of PD flagged; specificity is the fraction of healthy correctly cleared.

Q: How does audio get from browser to model? MediaRecorder → Web Audio decode → mono 16 kHz WAV → upload → quality gates (steady segment ≥ 1.5 s, SNR ≥ 15 dB) → Praat features + 28 spectral features (0–4 kHz) → models → stored record → report.

Q: Why OpenStreetMap? It is free, open (ODbL), has no API key, and covers India. Nominatim is used with a healthcare-type filter because the public Overpass servers were overloaded. The app follows OSM usage policy (User-Agent, 1 request/s, caching).

Q: Biggest limitation / future work? Small, non-Indian training data. Future work: collect consented Indian recordings (multiple languages, smartphones), prospective clinical validation with neurologist diagnosis, repeated recordings per person (average over takes), calibrated RPDE/DFA/PPE, same-person voice + gait data, and on-device inference for privacy.

Q: What did you add beyond the original project? See §6: browser audio conversion; extractor fixes; quality gates; two new screening models with tiers; new datasets (Sakar, Iyer); age-confound handling; leakage audit; formal report/PDF; delete; Progress; Indian care finder with map; recording guide with mic check; branding/UI; and tests (31 + 21).

19. Research paper guidance

Suggested title: “Smartphone-Grade Voice Screening for Parkinson's Disease: Cross-Dataset Evaluation, Age-Confound Control and a Deployed Web Portal for India”.

Defensible contributions

1. Evidence that classic jitter/shimmer/HNR features generalise across *lab* datasets (AUC 0.83 / 0.73) but fail on *phone* audio processed by an open pipeline (AUC 0.475).
2. Spectral/cepstral features (MFCC variability, CPPS, pitch SD) restore discrimination on phone audio (AUC 0.78; 0.77 age-matched) with explicit age-confound control.
3. A three-tier decision scheme with per-tier error rates for responsible screening.
4. A leakage audit showing how a random split inflates multimodal regression (R^2 0.91 → 0.25).
5. An open, reproducible end-to-end system (quality gating, reporting, care discovery for India).

Suggested structure: Abstract · Introduction (PD burden, voice biomarkers, access in India) · Related work (Little 2009, Tsanas 2010, Sakar 2019, Iyer 2023) · Data (Table §10) · Methods (pipeline §7, features §8, models §9, protocols, age weighting, tiers) · Results (Tables §11.3, §11.4, §11.5, §11.6) · System (architecture §4) · Discussion (channel effects, confounds, leakage) · Limitations (§17) · Conclusion.

Recommended figures/tables: ROC curves per model; tier distribution bar chart; age-confound table; leakage table; architecture diagram; report screenshot.

Claims to avoid: “diagnoses Parkinson's”, “90% accuracy”, “multimodal fusion improves prediction” (not supported after the leakage audit), “clinically validated”.

20. Glossary

Term	Meaning
PD	Parkinson's disease
UPDRS / MDS-UPDRS	(Movement Disorder Society) Unified Parkinson's Disease Rating Scale; Part III = motor exam
Sustained phonation	Holding a vowel (“aaah”) steadily
f0	Fundamental frequency (pitch)
MFCC	Mel-frequency cepstral coefficients (spectral envelope / timbre)
CPPS	Cepstral peak prominence, smoothed
LTAS	Long-term average spectrum
SNR	Signal-to-noise ratio
AUC / ROC	Area under the receiver-operating-characteristic curve
Grouped CV	Cross-validation where all recordings of a person stay in the same fold
Leakage	Test information (same person / target) influencing training
Confound	A variable (e.g. age) related to both input and label that can fake accuracy

Term	Meaning
Inverse-probability weighting	Re-weighting samples so classes share a covariate distribution
ODbL	Open Database Licence (OpenStreetMap)

21. References

1. Little MA, McSharry PE, Roberts SJ, Costello DAE, Moroz IM. Exploiting nonlinear recurrence and fractal scaling properties for voice disorder detection. *BioMedical Engineering OnLine*, 2007; 6:23.
2. Little MA, McSharry PE, Hunter EJ, Spielman J, Ramig LO. Suitability of dysphonia measurements for telemonitoring of Parkinson's disease. *IEEE Transactions on Biomedical Engineering*, 2009; 56(4):1015–1022.
3. Tsanas A, Little MA, McSharry PE, Ramig LO. Accurate telemonitoring of Parkinson's disease progression by noninvasive speech tests. *IEEE Transactions on Biomedical Engineering*, 2010; 57(4):884–893.
4. Sakar CO, Serbes G, Gunduz A, et al. A comparative analysis of speech signal processing algorithms for Parkinson's disease classification and the use of the tunable Q-factor wavelet transform. *Applied Soft Computing*, 2019; 74:255–263. (UCI dataset 470, CC BY 4.0)
5. Iyer A, et al. A machine learning method to process voice samples for identification of Parkinson's disease. *Scientific Reports*, 2023. Data: figshare doi:10.6084/m9.figshare.23849127 (CC BY 4.0).
6. Goldberger AL, et al. PhysioBank, PhysioToolkit, and PhysioNet. *Circulation*, 2000; 101(23):e215–e220. Dataset: “Gait in Parkinson's Disease” (Hausdorff et al.).
7. Martínez-Martín P, Rodríguez-Blázquez C, et al. Parkinson's disease severity levels and MDS-Unified Parkinson's Disease Rating Scale. *Parkinsonism & Related Disorders*, 2015; 21(1):50–54.
8. Boersma P, Weenink D. *Praat: doing phonetics by computer* (software).
9. Jadoul Y, Thompson B, de Boer B. Introducing Parselmouth: A Python interface to Praat. *Journal of Phonetics*, 2018; 71:1–15.
10. Hillenbrand J, Houde RA. Acoustic correlates of breathy vocal quality: dysphonic voices and continuous speech. *Journal of Speech and Hearing Research*, 1996; 39:311–321.
11. OpenStreetMap contributors. OpenStreetMap data, Open Database Licence (ODbL); Nominatim and Overpass API usage policies.

Check each reference's exact page and article numbers against the publisher before submitting the paper.